

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Modular Unit 1

Adaptive Classified

OL-T31 · Coordinate Geometry

Questions

Student book · questions with writing space

■ 12 pages

FAMILY MAP

Coordinate Geometry

F01	Determine coordinates from geometric information	6 Q 28 marks
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F02	Use the midpoint relationship	1 Q 2 marks
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F03	Read and plot Cartesian coordinates	3 Q 13 marks
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Question 28

4MA1-2H Q24

MODULAR UNIT 1

6 marks

24 $ABCD$ is a kite with $AB = AD$ and $CB = CD$

A is the point with coordinates $(-2, 10)$

B is the point with coordinates $\left(-\frac{27}{5}, 4\right)$

C is the point with coordinates $(4, -5)$

Work out the coordinates of D

(..... ,)

Question 28

4MA1-2H Q22 Q22

MODULAR UNIT 1

7 marks

- 22 $ABCD$ is a kite, with diagonals AC and BD , drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

A is the point with coordinates $(-3, 4)$

The diagonals of the kite intersect at the point M with coordinates $(0, 2)$

Given that $AB = AD = 6.5$ cm and the x coordinate of B is positive,

find the coordinates of the points B and D .

(.....,)

(.....,)

- 1 The point A has coordinates $(5, -4)$
The point B has coordinates $(13, 1)$

(a) Work out the coordinates of the midpoint of AB .

(.....,)

Find the value of j and the value of k .

 $k =$

18 A rectangle $ABCD$ is to be drawn on a centimetre grid such that

A has coordinates $(-4, -2)$

B has coordinates $(1, 10)$

C has coordinates $(19, a)$

D has coordinates (b, c)

(a) Work out the value of a , the value of b and the value of c .

$a =$

$b =$

$c =$

(4)

(b) Calculate the perimeter, in centimetres, of rectangle $ABCD$.

..... cm

(3)

Find the exact coordinates of the point where the line through B and D intersects the y -axis.

$$AB = AD \text{ and } CB = CD$$

The point D has coordinates $(8, 7)$

The straight line $\mathbf{L}$ passes through the points B and D

The straight line **L** is parallel to the line with equation $5y - 3x = 6$

Find an equation of AC

Give your answer in the form $px + qy = r$ where p , q and r are integers.

Show your working clearly.

(Total for Question 23 is 6 marks)

6 The points A and B are on a coordinate grid.

The coordinates of A are $(6, 4)$

The coordinates of B are $(17, j)$ where j is a constant.

The midpoint of AB has coordinates $(k, 15)$ where k is a constant.

Find the value of j and the value of k

$j = \dots\dots\dots$

$k = \dots\dots\dots$

(Total for Question 6 is 3 marks)

20 The straight line **L** has equation $y = 4x + 7$

The straight line **M** is perpendicular to **L** and passes through the point with coordinates (8, 1)

Find an equation for **M**

Give your answer in the form $y = mx + c$

(Total for Question 20 is 3 marks)

- 1 The point P has coordinates $(3, 4)$
The point Q has coordinates $(9, 16)$

M is the midpoint of the line PQ

Find the coordinates of M

(..... ,)

(Total for Question 1 is 2 marks)